



# Cambridge International AS & A Level

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## FURTHER MATHEMATICS

9231/44

Paper 4 Further Probability &amp; Statistics

October/November 2025

1 hour 30 minutes

You must answer on the question paper.

You will need: List of formulae (MF19)

### INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.

### INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **12** pages. Any blank pages are indicated.

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- 1 Local residents are concerned about the speed of cars travelling through their village. They record the speed, in  $\text{km h}^{-1}$ , of a random sample of 13 cars travelling through their village. The recorded speeds are as follows.

40 53 59 42 43 48 62 67 46 82 66 45 70

- (a) Construct a 90% confidence interval for the population mean speed of cars passing through the village. [5]

$$C.I. = (\bar{x}) \pm \frac{t_{\alpha/2, n-1}}{2} \sqrt{\frac{S_{xx}}{n}}$$

$$\sum x = 40 + 53 + 59 + 42 + 43 + \dots + 70 = 723$$

$$\sum x^2 = 42261$$

$$\bar{x} = \sum x / n = 723 / 13 = 55.6$$

$$S_{xx}^2 = \frac{1}{n-1} \left( \sum x^2 - \frac{(\sum x)^2}{n} \right) = \frac{1}{12} \left( 42261 - \frac{723^2}{13} \right)$$

$$= 170.92$$

$$C.I. = 55.6 \pm t_{0.05, 12} \sqrt{\frac{170.92}{13}}$$

$$= 55.6 \pm 1.782 (3.626)$$

$$= 55.6 \pm 6.46$$

$$= [49.2, 62.1]$$

- (b) State an assumption that is necessary for the confidence interval found in 1(a) to be valid. [1]

underlying population distribution should be normally distributed



- 2 The continuous random variable  $X$  has probability density function  $f$  given by

$$f(x) = \begin{cases} \frac{4}{9}(x+1) & 0 \leq x \leq 1, \\ (x-2)^2 & 1 < x \leq 2, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Find the cumulative distribution function of  $X$ .

[3]

$$F(x) = \int f(x) dx$$

$$\int_0^x \frac{4}{9}(t+1) dt \rightarrow \left[ \frac{4}{18}t^2 + \frac{4}{9}t \right]_0^x$$

$$\rightarrow \frac{4}{18}x^2 + \frac{4}{9}x$$

$$\int_1^x (t-2)^2 dt \rightarrow \frac{1}{3}(t-2)^3 - \frac{1}{3}(-3)$$

$$\rightarrow \frac{1}{3}(x-2)^3 + 1$$

$$F(x) = \begin{cases} 0 & x < 0 \\ \frac{2}{9}x^2 + \frac{4}{9}x & 0 \leq x < 1 \\ \frac{1}{3}(x-2)^3 + 1 & 1 \leq x \leq 2 \\ 1 & x > 2 \end{cases}$$

- (b) Find the exact value of the median of  $X$ .

[2]

$$\frac{2}{9}m^2 + \frac{4}{9}m = 0.5$$

$0 \leq x \leq 1 \rightarrow$  correct interval

$$4m^2 + 8m - 9 = 0$$

$0 \leq p \leq \frac{2}{3} \rightarrow$  so  $p = 0.5$  in the interval

$$\frac{-8 \pm \sqrt{8^2 + 144}}{8}$$

$$= \frac{-8 \pm 4\sqrt{5}}{8}$$

$$= \frac{-8 + 4\sqrt{5}}{8}$$

The random variable  $Y$  is defined by  $Y = \sqrt{X}$ .

- (c) Find the cumulative distribution function of  $Y$ .

[3]

limits  $\rightarrow 0 \leq Y \leq 1$

$\rightarrow \sqrt{1} \leq y \leq \sqrt{2}$

$$F(y) = \begin{cases} 0 & y < 0 \\ \frac{2}{9}(y^2)^2 + \frac{4}{9}y^2 & 0 \leq y \leq 1 \\ \frac{1}{3}(y^2 - 2)^3 + 1 & 1 \leq y \leq \sqrt{2} \\ 1 & y > \sqrt{2} \end{cases}$$

substitute  $Y^2 = X$

- (d) Determine whether the median of  $Y$  is greater than, or less than, the median of  $X$ .

[2]

$$\frac{2}{9}y^4 + \frac{4}{9}y^2 = 0.5 \quad z = y^2$$

$$z = \frac{-2 + \sqrt{13}}{2} \quad \text{hence } y^2 = \frac{-2 + \sqrt{13}}{2}$$

$$\sqrt{\frac{-2 + \sqrt{13}}{2}} > \frac{-2 + \sqrt{13}}{2}$$

$$\boxed{\text{med}(Y) > \text{med}(X)}$$



- 3 Two different types of juice extractor, machine  $X$  and machine  $Y$ , are being compared. The manufacturer claims that machine  $Y$  extracts more juice per orange on average than machine  $X$ .

A random sample of 20 oranges is selected and each carefully measured. Oranges of a similar size and mass are then paired and numbered from 1 to 10. One orange from each pair is randomly allocated to machine  $X$  with the other allocated to machine  $Y$ . The amount of juice, in ml, extracted from each orange is recorded in the following table.

|             | pair |    |    |    |    |    |    |    |    |    |
|-------------|------|----|----|----|----|----|----|----|----|----|
|             | 1    | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 |
| machine $X$ | 65   | 73 | 58 | 61 | 72 | 79 | 64 | 65 | 69 | 71 |
| machine $Y$ | 68   | 72 | 64 | 63 | 75 | 82 | 63 | 63 | 72 | 74 |

- (a) Use a  $t$ -test to test the manufacturer's claim at the 1% significance level.

[7]

paired differences  $t$ -test

$$H_0: \mu_y = \mu_x \quad H_1: \mu_y > \mu_x$$

$$\sum d = (-3) + (1) + (-6) + \dots + (-3) = -19 \text{ or } 19$$

$$\sum d^2 = 91$$

$$s_d^2 \rightarrow \text{variance of differences} \quad \bar{d} = \frac{19}{10} = 1.9$$

$$= \frac{1}{n-1} \left( \sum x^2 - \frac{(\sum x)^2}{n} \right) = \frac{1}{9} \left( 91 - \frac{(19)^2}{10} \right)$$

$$= 6.1$$

$$T = \frac{\bar{d} - \mu_d}{\sqrt{\frac{s_d^2}{n}}} = \frac{1.9 - 0}{\sqrt{\frac{6.1}{10}}} = 2.43$$

critical value =  $t_{0.99, 9}$  <sup>one-tailed</sup> = 2.821

$$2.43 < 2.821 \rightarrow \text{so accept } H_0$$

$\rightarrow$  not sufficient enough evidence to support manufacturer's claim



(b) State an assumption required for the test in 3(a) to be valid.

[1]

underlying distribution of population differences is normally distributed

The manufacturer notices that the amount of juice extracted from the oranges in pair 4 were recorded incorrectly. In fact, the amount of juice extracted from machine  $X$  was 60ml and from machine  $Y$  was 62ml.

(c) Explain, with justification, whether the conclusion of the test in 3(a) remains the same.

[2]

no as the difference remains  $-2$  for pair 4, hence not affecting  $\sum d$  or  $\sum d^2$

→ So the conclusion remains the same as the test statistic remains the same



- 4 The random variable  $X$  takes values 1 and 2 with probabilities  $\frac{2}{5}$  and  $\frac{3}{5}$  respectively.

(a) Write down the probability generating function  $G_X(t)$  of  $X$ .

[1]

$$G_X(t) = \frac{2}{5}t + \frac{3}{5}t^2$$

$$\rightarrow t \left( \frac{2}{5} + \frac{3}{5}t \right)$$

The random variable  $Y$  is the sum of four independent observations of  $X$ .

(b) Find the probability generating function  $G_Y(t)$  of  $Y$ . Give your answer in the form  $G_Y(t) = at^m(b+ct)^n$ , where  $a, b, c, m$  and  $n$  are constants to be determined.

[2]

$$G_Y(t) = (G_X(t))^4$$

$$= t^4 \left( \frac{2}{5} + \frac{3}{5}t \right)^4$$

(c) Use  $G_Y(t)$  to find  $P(Y=6)$ .

[2]

$$P(Y=6) \rightarrow \text{find } t^6 \text{ term}$$

$$\left( \frac{2}{5} + \frac{3}{5}t \right)^4 \rightarrow \binom{4}{2} \times \left( \frac{3}{5}t \right)^2 \times \left( \frac{2}{5} \right)^2$$

$$= \frac{216}{625} t^2 \times t^4 = \frac{216}{625} t^6$$

$$P(Y=6) = \frac{216}{625}$$







$$G_Y(t) = t^4 \left( \frac{2}{5} + \frac{3}{5}t \right)^4$$

(d) Find  $\text{Var}(Y)$ .

[5]

$$E(Y) = G'_Y(1) \rightarrow \text{use product rule}$$

$$G'_Y(t) = 4t^3 \left( \frac{2}{5} + \frac{3}{5}t \right)^4 + 4t^4 \times \frac{3}{5} \left( \frac{2}{5} + \frac{3}{5}t \right)^3$$

$$G'_Y(1) = 4 + \frac{12}{5} = 6.4$$

$$G''_Y(t) = \text{double product rule}$$

$$\rightarrow 12t^2 \left( \frac{2}{5} + \frac{3}{5}t \right)^4 + 4t^3 \times \frac{3}{5} \times 4 \left( \frac{2}{5} + \frac{3}{5}t \right)^3$$

$$+ \frac{48}{5} t^3 \left( \frac{2}{5} + \frac{3}{5}t \right)^3 + \frac{12}{5} t^4 \times \frac{3}{5} \times 3 \left( \frac{2}{5} + \frac{3}{5}t \right)^2$$

$$G''_Y(1) = 12 + \frac{48}{5} + \frac{48}{5} + \frac{36 \times 3}{25} = \frac{888}{25}$$

$$\text{Var}(Y) = G''_Y(1) + G'_Y(1) - \left( G'_Y(1) \right)^2$$

$$= \frac{888}{25} + \frac{32}{5} - \left( \frac{32}{5} \right)^2 = \frac{124}{25}$$



- 5 A driving instructor believes that the performance (pass or fail) of students when taking a driving test is associated with their age. The following table summarises the number of students who pass and who fail, and the ages in years of the students taking the test, over a period of three years.

|       | age of student |       |         |       |
|-------|----------------|-------|---------|-------|
|       | under 20       | 20-40 | over 40 | total |
| pass  | 34             | 41    | 6       | 81    |
| fail  | 16             | 39    | 9       | 64    |
| total | 50             | 80    | 15      | 145   |

$H_0$   
: performance  
independent  
of age

$H_1$ : performance  
not independent of age

Test, at the 10% significance level, whether performance is independent of age of student.

[7]

table of expected values

|   | <20   | 20-40 | 40   |
|---|-------|-------|------|
| p | 27.93 | 44.69 | 8.38 |
| f | 22.07 | 35.31 | 6.62 |

table values

$$\frac{R_T \times C_T}{\text{Total}}$$

$$\text{Test statistic} = \sum \frac{(\text{Obs} - \text{Exp})^2}{\text{Exp}}$$

$$= \frac{(34 - 27.93)^2}{27.93} + \dots + \frac{(6.62 - 9)^2}{6.62}$$

$$= 1.319 + 0.305 + 0.676$$

$$+ 1.669 + 0.386 + 0.856$$

$$= 5.211$$

$$\text{Critical value } \chi^2_{(r-1)(c-1)}(0.9) = \chi^2_{(0.9)} = 4.605$$

$5.211 > 4.605 \rightarrow$  reject null hypothesis

$\rightarrow$  sufficient evidence to suggest age and performance are not independent



- 6 Students of the same age from two schools, school  $A$  and school  $B$ , take a large number of quizzes throughout the year and are each awarded a mark out of 1000. The marks of 123 students in school  $A$  and 147 students in school  $B$  are ranked from lowest (rank 1) to highest (rank 270). The sum of the ranks of the students from school  $A$  is 15355.

- (a) Carry out a Wilcoxon rank-sum test at the 5% significance level to investigate whether there is a difference in average marks between the students in school  $A$  and school  $B$ . [6]

$$H_0: \text{med}_A = \text{med}_B \quad H_1: \text{med}_A \neq \text{med}_B$$

$$W = \min(R_m, m(n+m+1) - R_m) \quad \text{where } m \leq n$$

$$= \min(15355, (123)(271) - 15355)$$

$$= \min(15355, 1978)$$

$$W = 15355$$

$$\mu, \sigma^2 = \left( \frac{1}{2}m(m+1), \frac{1}{12}m(m+1)(m+2) \right) \quad \text{using normal approx}$$

$$= (16666.5, 408329.25)$$

$$\text{Test statistic} = \frac{W - \mu + 0.5}{\sigma} = \frac{15355 - 16666.5 + 0.5}{\sqrt{408329.25}}$$

$$= -2.05$$

$$\text{Critical value} = z_{0.95} = -1.96 \quad \text{two-tailed} \quad \text{graph}$$

$$\text{as } -2.05 < -1.96 \rightarrow \text{reject } H_0$$

$\rightarrow$  sufficient evidence to suggest a difference in median marks

- (b) State an assumption that is required for the Wilcoxon rank-sum test to be valid. [1]

underlying distributions are symmetrical





## Additional page

If you use the following page to complete the answer to any question, the question number must be clearly shown.

[illegible]

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